Karnataka Residential Educational Institutions Society

ICT Class 8 - Summative Assessment 2

Summative Assessment (SA-2)

Class: Class 8

Assessment Type: Summative Assessment (SA-2)

Total Marks: 40

Duration: 2 hours

Topics Covered: Algorithms, Emerging Technologies, Cyber Safety, Computational Thinking

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Part A: MCQ (10 x 1 = 10 marks)

a. Which sorting algorithm has the best average-case time complexity?

b. Bubble Sort

c. Selection Sort

d. Merge Sort

e. Insertion Sort

b. What is the time complexity of binary search on a sorted array of `n` elements?

c. O(n)

d. O(n log n)

e. O(log n)

f. O(1)

c. Which of the following is NOT a characteristics of a good algorithm?

d. Finiteness

e. Ambiguity

f. Definiteness

g. Effectiveness

d. What does IoT stand for?

e. Internet of Technologies

f. Internet of Things

g. Integration of Things

h. Interface of Technology

e. Which technology is the foundation of cryptocurrency?

f. Cloud Computing

g. Artificial Intelligence

h. Blockchain

i. Big Data

f. What is the primary purpose of a flowchart?

g. To write code
h. To represent an algorithm visually
i. To compile programs
j. To store data
g. Which of these is a characteristic of cloud computing?
h. Local storage only
i. On-demand self-service
j. Single-user access
k. Fixed capacity
h. In pseudocode, which symbol represents a decision/condition?
i. Rectangle
j. Diamond
k. Oval
l. Parallelogram
i. What is machine learning?
j. A type of hardware
k. A subset of AI that enables systems to learn from data
l. A programming language
m. A database management system
j. Which cyber safety practice is most important?
k. Sharing passwords with friends
l. Using public Wi-Fi for banking
m. Enabling two-factor authentication
n. Clicking on unknown email links

Part B: Short Answer (8 x 2 = 16 marks)

a. Write an algorithm (in pseudocode) to find the largest element in an unsorted list of `n` numbers. Analyse its time complexity.
b. Differentiate between **artificial intelligence**, **machine learning**, and **deep learning** with one example of each.
c. Explain the concept of **divide and conquer** with an example. How does Merge Sort use this strategy?
d. What is **Big Data**? Explain the 5 V's of Big Data with one sentence each.
e. Write a Python program that implements **linear search** and **binary search** on a list. Compare their performance.
f. Explain three cyber safety measures that every student should follow while using the internet. Why is each important?

g. What is **cloud computing**? Differentiate between IaaS, PaaS, and SaaS with one real-life example each.

h. Write the pseudocode for a **bubble sort** algorithm. Trace it on the list `[5, 3, 8, 1, 2]` showing each pass.

Part C: Long Answer (4 x 3 = 12 marks)

a. Write a complete Python program that implements **Insertion Sort**. Trace the algorithm on the list `[12, 11, 13, 5, 6]` showing each step. Analyse the time complexity for best, average, and worst cases.

b. Explain the following emerging technologies with detailed examples:

- **Blockchain**: Structure, how it works, and one application beyond cryptocurrency
- **Internet of Things (IoT)**: Components, working, and one smart city application
- **Artificial Intelligence**: Two real-world applications in education and healthcare

c. Design a **traffic management system** using algorithms. Write:

- The problem statement
- An algorithm (in pseudocode) that assigns green signal duration based on vehicle density
- A flowchart description for the decision-making process
- How IoT sensors could enhance this system

d. A school wants to digitise its library system. Write:

- A detailed algorithm to search for a book by title, author, or ISBN
- The database design (tables with columns) using SQL
- Python pseudocode for the search function with error handling
- How cloud computing could be used to make the system accessible from anywhere

Part D: Application Based (2 x 1 = 2 marks)

a. A social media company wants to analyse user behaviour. Explain how:

- Big Data analytics can help understand user preferences
- Machine Learning can improve content recommendations
- AI can detect and remove harmful content automatically

b. A farmer wants to use technology to improve crop yield. Explain how:

- IoT sensors can monitor soil moisture and temperature
- Data analytics can predict optimal planting times
- Cloud computing can help access weather data and market prices remotely